

Subrat Prasad Panda

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🏠 Scholar

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Research Statement: I am broadly interested in **Trustworthy Reinforcement Learning**, particularly in designing interpretable, robust, and verifiable policies for safety-critical domains. My current work focuses on **Neurosymbolic** approaches such as programmatic policies, combining explicit knowledge with learning to bridge System 1 (neural) and System 2 (symbolic) reasoning.

Education

- 2023 – 2027 📖 **Ph.D., Computer Science**, NTU Singapore
Thesis: *Neurosymbolic Approaches for Trustworthy Reinforcement Learning*
Advisors: Arvind Easwaran (NTU), Blaise Genest (IPAL, CNRS France & Singapore)
- 2019 – 2021 📖 **M.Tech., Computer Science**, ISI Kolkata, India
Thesis: *Strategies for User Allocation and Service Placement in Multi-Access Edge Computing*
CPI: 82.1% Advisor: Ansuman Banerjee
- 2012 – 2016 📖 **B.Tech., Electronics and Instrumentation**, NIT Silchar, India
Thesis: *Real-Time Position Control of DC Motor using PID and Fuzzy Logic*
CPI: 7.92/10 Advisor: Munmun Khanra

Employment History

- 2022 – 2023 📖 **Research Associate**, CNRS@CREATE, Singapore
• Designed a granular control system for microgrids using Deep Reinforcement Learning.
- 2021 – 2022 📖 **Machine Learning Engineer**, InFocusP Innovation Pvt. Ltd., Pune, India
• Developed solutions for the Abstraction and Reasoning Challenge (ARC) using Reinforcement Learning. Under attachment to Google X (USA).
- 2021 – 2021 📖 **Research Engineer**, TiHAN, IIT Hyderabad
• Developed braking and throttle control systems using Arduino for a test vehicle.
• Deployed a lane detection model on the NVIDIA AGX board.
- 2016 – 2018 📖 **Aviation Officer**, HPCL, Chandigarh International Airport
• Managed maintenance and repair of equipment for aircraft refueling operations.
- 2015 – 2015 📖 **Summer Intern**, DRDO, New Delhi, India
• Enhanced colored image contrast in the spatial domain using histogram equalization and cubic spline interpolation.

Research Publications

Conference Proceedings

- 1 **S. P. Panda**, B. Genest, and A. Easwaran, “Neurosymbolic learning of interpretable abstraction in reinforcement learning,” Work-in-Progress, 2027.
- 2 **S. P. Panda**, L. Yuke, B. Genest, D. Zikelic, and A. Easwaran, “Learning unions of polytopes as abstractions and barriers for the verification of rl with discontinuities,” Work-in-Progress, 2027.
- 3 **S. P. Panda**, B. Genest, and A. Easwaran, “Reasoning with incremental knowledge for sample efficient hierarchical reinforcement learning,” ECML-PKDD 2026, 2026.
- 4 **S. P. Panda**, B. Genest, A. Easwaran, and P. N. Suganthan, “Vanilla gradient descent for oblique decision trees,” in *ECAI 2024*, IOS Press, Oct. 2024. 🌐 DOI: 10.3233/faia240607

- 5 **S. P. Panda**, A. Banerjee, and A. Bhattacharya, "User allocation in mobile edge computing: A deep reinforcement learning approach," in *2021 IEEE International Conference on Web Services (ICWS)*, IEEE, 2021, pp. 447–458.
- 6 **S. P. Panda**, K. Ray, and A. Banerjee, "Service allocation/placement in multi-access edge computing with workload fluctuations," in *International Conference on Service-Oriented Computing*, Springer, 2021, pp. 747–755.
- 7 **S. P. Panda**, K. Ray, and A. Banerjee, "Dynamic edge user allocation with user specified qos preferences," in *International Conference on Service-Oriented Computing*, Springer, 2020, pp. 187–197.

Journal Articles

- 1 **S. P. Panda**, B. Genest, and A. Easwaran, "Approximation-free oblique decision trees," *JMLR*, 2026.
- 2 **S. P. Panda**, B. Genest, A. Easwaran, R. Rigo-Mariani, and P. Lin, "Methods for mitigating uncertainty in real-time operations of a connected microgrid," *Sustainable Energy, Grids and Networks*, vol. 38, p. 101334, 2024.

Workshops

- 1 **S. P. Panda**, B. Genest, and A. Easwaran, *Programmatic reinforcement learning for trustworthy microgrid management*, AI4X 2025 International Conference, 2025.

Interests and Skills

Research	Reinforcement Learning, Neuro-Symbolic AI, Knowledge and Reasoning
Programming	Python, PyTorch, MATLAB, $\LaTeX$, ROS
Courses	Mathematical Statistics, Deep Learning, Machine Learning, Computer Vision

Miscellaneous Experience

Awards and Achievements

- 2023 **DesCartes Scholarship** — Ph.D. scholarship awarded by CNRS@CREATE under the DesCartes project.
- 2021 **M.Tech Best Thesis Nomination** — Ranked among the top 5 theses of the batch.
- 2019 **M.Tech Scholarship** — Scholarship awarded to pursue a master's degree.
- 2016 **Top 25 Finalist**, General Electric Edition Challenge — IoT-based engine exhaust gas monitoring system.
- 2015 **1st Round**, Texas Instruments Innovation Challenge — Non-invasive digital blood pressure meter.

Academic Activities and Mentoring

Reviewer	Reviewer for CAV'26, AAAI'26, ECAI'26, IEEE TSMC, RTSS'26, and NeurIPS'26.
F1Tenth	F1Tenth Racing Team , NTU — Advised students on control and navigation strategies.
FYP	Undergraduate Final Year Projects , NTU — Guided projects on CNN robustness and programmatic policies using LLMs.
URECA	Undergraduate Research , NTU — Mentored projects on Decision Transformers, Mechanistic Interpretability, ARC Reasoning, and Physics-Inspired RL.
Leadership	Vice President , CCDS Student Club - Played a key role in organizing events such as the Deep Learning Bootcamp and AI Meetup.

References

Arvind Easwaran

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